

Name: _____

A Level Mathematics : Pure Mathematics

Topic 2 :Algebra and functions -Composite and inverse functions

Date:

Time:

Total marks available:

Total marks achieved: _____

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Questions

Q1.

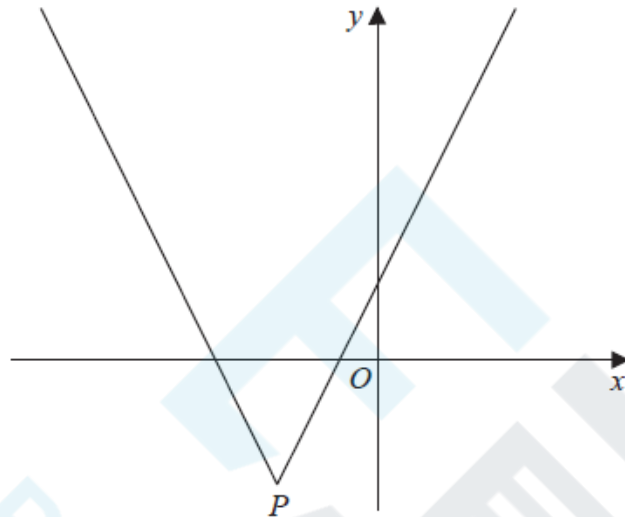


Figure 2

Figure 2 shows a sketch of the graph with equation

$$y = 2|x + 4| - 5$$

The vertex of the graph is at the point P , shown in Figure 2.

(a) Find the coordinates of P .

(2)

(b) Solve the equation

$$3x + 40 = 2|x + 4| - 5$$

(2)

A line l has equation $y = ax$, where a is a constant.

Given that l intersects $y = 2|x + 4| - 5$ at least once,

(c) find the range of possible values of a , writing your answer in set notation.

(3)

(Total for question = 7 marks)

Q2.

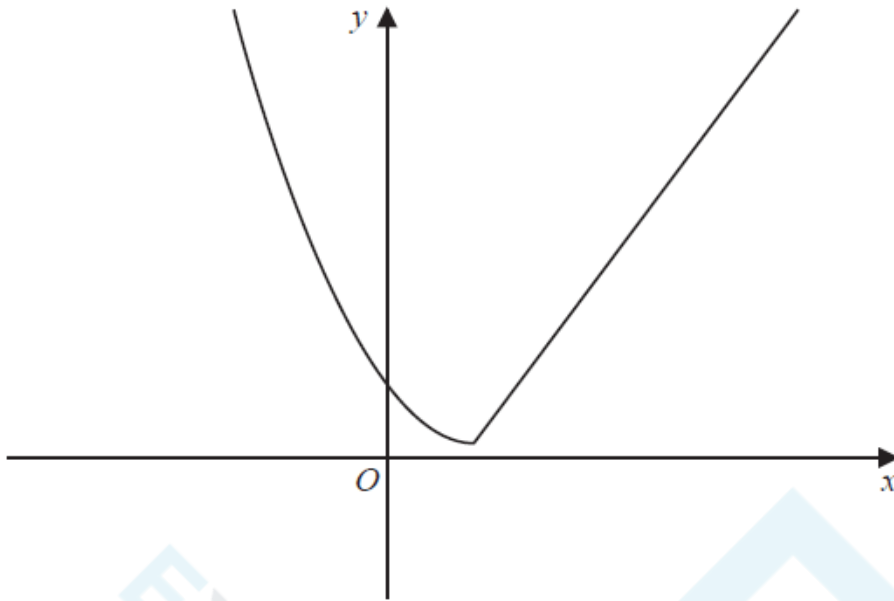


Figure 4

Figure 4 shows a sketch of the graph of $y = g(x)$, where

$$g(x) = \begin{cases} (x-2)^2 + 1 & x \leq 2 \\ 4x - 7 & x > 2 \end{cases}$$

(a) Find the value of $gg(0)$.

(2)

(b) Find all values of x for which

$$g(x) > 28$$

(4)

The function h is defined by

$$h(x) = (x-2)^2 + 1 \quad x \leq 2$$

(c) Explain why h has an inverse but g does not.

(1)

(d) Solve the equation

$$h^{-1}(x) = -\frac{1}{2}$$

(3)

(Total for question = 10 marks)

Q3.

Given

$$f(x) = e^x, \quad x \in \mathbb{R}$$

$$g(x) = 3 \ln x, \quad x > 0, x \in \mathbb{R}$$

(a) find an expression for $gf(x)$, simplifying your answer.

(2)

(b) Show that there is only one real value of x for which $gf(x) = fg(x)$

(3)

(Total for question = 5 marks)

Q4.

$$g(x) = \frac{2x+5}{x-3} \quad x \geq 5$$

(a) Find $gg(5)$.

(2)

(b) State the range of g .

(1)

(c) Find $g^{-1}(x)$, stating its domain.

(3)

(Total for question = 6 marks)

Q5.

The curve with equation $y = 3 \times 2^x$ meets the curve with equation $y = 15 - 2^{x+1}$ at the point P .

Find, using algebra, the exact x coordinate of P .

(Total for question = 4 marks)